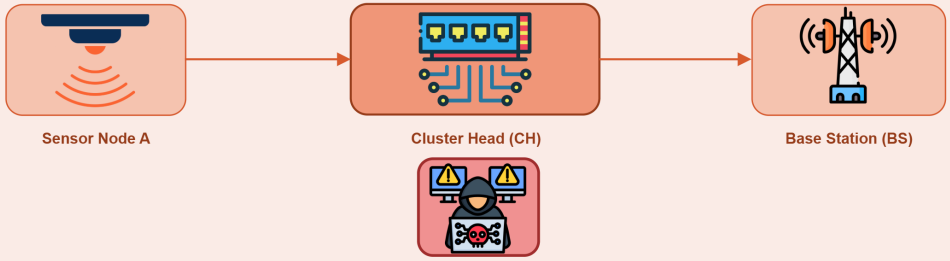
Prior WSN-IDS Research

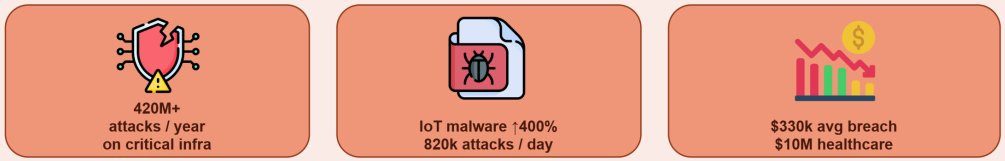
What the field built and what it left out

① The deployment environment

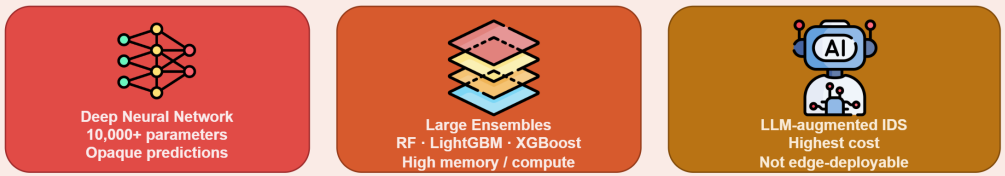


Blackhole | Grayhole | Flooding | TDMA Disruption

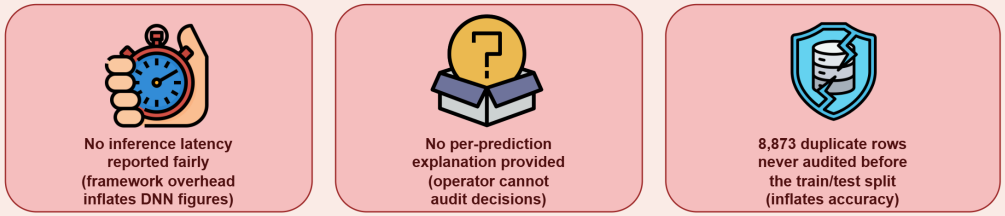
② The stakes: why this matters now



③ What prior work reached for (and why it falls short)



④ Three gaps left open



Research Question

Can one small, readable, tuned Decision Tree match deep models on WSN-DS, explain every prediction with SHAP, and report its own limits honestly?

Result: high accuracy reported, low trustworthiness, not edge-deployable

This work →

This Work

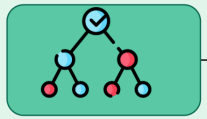
Audit first · tune smart · explain every call

① Data-quality audit before any modelling



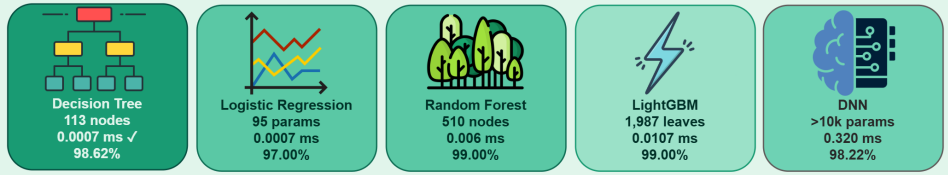
374,661 raw records checked for exact duplication
8,873 rows (2.37%) removed — concentrated in Normal + Grayhole classes
Remaining: 365,788 clean rows · stratified 70/30 split, seed 42

② GridSearchCV Decision Tree — interpretability-constrained

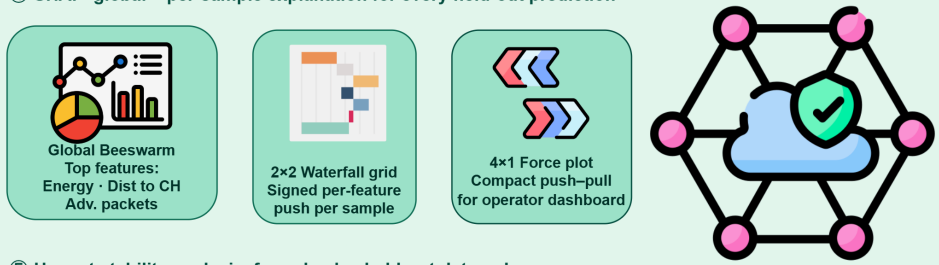


GridSearchCV over max_depth ∈ {3,5,7} · min_samples_split ∈ {2,5,10}
Selected: depth=7, min_samples_split=10 → 113 nodes
Accuracy 98.62% · Weighted F1 0.99 · 0.0007 ms / sample

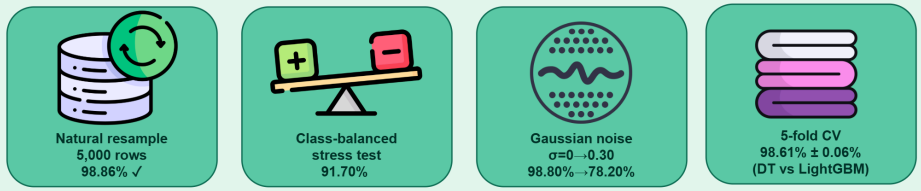
③ Fair latency benchmark same warm-up protocol for all 5 models



④ SHAP global + per-sample explanation for every held-out prediction



⑤ Honest stability analysis four checks, held-out data only



Accurate · Lightweight · Audited · Fully Explained · Honestly Scoped